

CEO Age, CEO Tenure, and Corporate Cash Holdings: The Moderating Effect of CEO Internalization¹

(Received Feb 29, 2024; First Revision May 02, 2024; Second Revision Oct 29, 2024;
Accepted Nov 18, 2024)

Purpose – The aim of this study is to examine the impacts of CEO age and CEO tenure on the cash holdings of firms.

Design/methodology/approach – To address the endogeneity of CEO age and tenure, we use a two-stage panel regression model (2SLS) using the instrumental variables for running the regression of CEO age or/and tenure in the first stage.

Findings – There is a negative association between CEO age and cash holdings, and a positive correlation between CEO tenure and cash holdings. The effects of observation period and threshold tenure on cash holdings are all significant and positive, indicating that CEOs deliberately obscure their high risk-taking attitude during the initial observation periods and when a CEO's tenure approaches tenure threshold. CEO internalization has a negative moderating effect on the observation period but no significant effect on the tenure threshold.

Research limitations/implications – Under the Personal Data Protection Law in Taiwan, it is not necessary to disclose the age of a CEO. Through various channels and over more than three years, we managed to gather 88.74% of the data on CEOs' ages and tenures.

Practical implications/Social implications – Our findings suggest that CEO age and tenure can predict cash holdings. The internalization of CEOs with minimal supervision moderates the effects of CEO age and tenure. We recommend stronger investor protection through external oversight and stricter corporate governance.

Originality/value – The main contribution of our study is the collection of CEO age and tenure data through manual methods addressing a gap in research on this topic. To the best of our knowledge, this is the first study to examine the effects of CEO age and CEO tenure on cash holdings from listed firms in Taiwan for the period 2010 to 2019.

Keywords – CEO age, CEO tenure, CEO internalization, Observation period effect, Tenure threshold effect

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¹ The authors would like to thank Kili C. Wang (the editor) and one anonymous referee for valuable comments that greatly improved this paper. They would like to thank Vincent Y.L. Chang at the 2023 TRIA for valuable comments.

DOI:10.6656/MR.202501_44(1).ENG002

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CEO 年齡、CEO 任期與企業現金持有：CEO 內部化的調節效果²

(Received Feb 29, 2024; First Revision May 02, 2024; Second Revision Oct 29, 2024;
Accepted Nov 18, 2024)

研究目的：CEO 年齡與任期對企業現金持有的影響。

研究設計/方法：使用兩階段迴歸模型 (2SLS) 以解決內生性問題。

研究結果：CEO 年齡 (任期) 與現金持有存在負 (正) 相關，觀察期效果和任期門檻對現金持有的影響為正，CEO 內部化對觀察期效果有負面調節作用，對任期門檻無影響。

研究限制/啟發：台灣個人資料保護法規定不強制公開 CEO 年齡，我們花超過三年以上透過各種管道，收集到 88.74% 的 CEO 年齡與任期數據。

理論/實務/社會意涵：CEO 年齡和任期可預測現金持有，監管較少的內部化 CEO 可調節 CEO 年齡與任期的影響，建議透過外部監督與更嚴格公司治理加強投資者保護。

創見/價值：手動收集 CEO 年齡和任期數據，填補了此領域研究的空白，據我們理解，此篇是首篇探討台灣上市公司 CEO 年齡與任期對現金持有影響研究。

關鍵字：CEO 年齡、CEO 任期、CEO 內部化、觀察期效應、任期門檻效應

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² 我們謹此感謝主編汪琪玲教授在本文投稿和審查期間的協助，以及審稿人的寶貴意見，為本文提供了重要的見解。我們也非常感謝張永郎教授在 2023 年台灣風險與保險學會 (TRIA) 年會暨學術研討會所提供的意見。

1. Introduction

Corporate financial decisions are made by the CEO. The upper echelons theory (UET), proposed by Hambrick and Mason (1984), explains how the personal characteristics of CEOs, such as cognitive abilities and values, affect corporate policies based on the assumption of limited rationality. CEO age and tenure are the most frequently used variables of personal risk attitude (Chen & Zheng, 2014; Hambrick & Mason, 1984). One stream of literature examines the impacts of CEO characteristics. For example, corporate decision making (Cline & Yore, 2016; Simsek, 2007), corporate risk (Chowdhury & Fink, 2017; Farag & Mallin, 2018; Serfling, 2014), and cash holdings decisions (Gaio et al., 2022; Guha & Rahim, 2019; Lim & Lee, 2019; Orens & Reheul, 2013; Serfling, 2014; Suherman et al., 2021)³. Another stream of literature examines how much cash a company should hold based on corporate cash management policies⁴. Most of the literature discusses developed countries. In contrast, there are very few studies of emerging economies. The influence of CEO characteristics on financial decision-making might differ between developed countries and emerging economies. Therefore, it is interesting to verify such relationships to provide corporate management policy suggestions.

Cash is the safest of all assets due to its high liquidity. Decision makers can modify corporate risk via cash holdings and reduce the risk of uncertainty due to lack of liquidity rather than focus on earnings rewards (Almeida et al., 2004). Looking back on some of the major financial events, one of the key factors in weathering such events is sufficient cash holdings (Campello et al., 2011; Duchin et al., 2010). Lins et al. (2010) report that activities that management of cash holdings is of great importance to the financial health of companies. The more cash a company holds the lower its risk of liquidity failure due to unanticipated spending needs (Ferreira & Vilela, 2004; Opler et al., 1999), lack of opportunities for external financing, and transaction costs of financing (Al-Najjar & Belghitar, 2011). However, some studies indicate that the higher the cash holdings, the higher the opportunity costs for the firm (Pinkowitz et al., 2006), the higher the probability of overinvestment (Richardson, 2006), and the lower the potential tax shield for debt financing (Bigelli & Sánchez-Vidal, 2012).

This study explores the effects of CEO characteristics, especially CEO age and tenure, on corporate cash holdings. In general, age and tenure are two important demographic variables. Age is associated with accumulating life experience and expected periods for each life stage now and into the future. Cognitive abilities and values change as life experience accumulates. CEO tenure represents the time a CEO holds that position in a company. The longer the tenure, the more experience a CEO accumulates in terms of business operations. Both CEO age and tenure are time dependent. With increases in age and tenure, CEOs develop different risk attitudes (Malmendier et al., 2011).

UET states that the heterogeneity of managers' traits creates unique management styles. CEO age and tenure are the most frequently used variables of CEO traits. In this study, unique, hand-collected, individual CEO data from listed firms in Taiwan from 2010 to 2019 are used to analyze the impact of CEO age and tenure on cash holdings. Our results show a significant adverse effect of CEO age, consistent with Guha and Rahim (2019) and Mun et al. (2020). The longer the CEO tenure, the higher the level of cash holdings, consistent with Orens and Reheul (2013). In addition, the interaction effects between CEO age or CEO tenure and CEO internalization⁵ indicate that CEO internalization mitigates the negative effect of CEO age and increases the positive effect of CEO tenure. We further examine the impact of CEO observation and tenure threshold periods on cash holdings. Our results indicate that a new CEO is associated with higher cash holdings during the observation period. There is also a positive impact of CEO tenure threshold on cash holdings, consistent with the findings of a previous study (Hu, Kale, et al., 2011). In terms of the interaction effects, internalized CEO decreases the positive effect of CEO observation period but does not influence the effect of CEO tenure threshold on cash holdings.

Past literature on cash holdings, such as Dittmar et al. (2003), concurs with the present study that agency problems are a significant determinant of corporate cash holdings. However, the analysis by Dittmar et al. (2003) is conducted from a national perspective through cross-country comparisons, rather than from the standpoint of CEO demographic characteristics. Hu et al. (2019) analyzed the impact of CEOs' early-life poverty experiences on their cash policies, incorporating CEO age as a control variable in their model, and found no significant relationship between CEO age and cash holdings. Kuan et al. (2011) investigated the link between corporate governance and cash policies in family-controlled firms. Lim and Lee (2019) examined the relationship between excess cash holdings and CEO characteristics in non-financial Korean listed companies, employing CEO characteristics such as ownership type, presence of stock options, inclusion in a chaebol, and CEO tenure, but not including CEO age. This study, in contrast, simultaneously

³ Some studies also focus on characteristics such as gender (Zeng & Wang, 2015), CEO early life experience (Hu et al., 2019), and education (Mun et al., 2020; Orens & Reheul, 2013).

⁴ Examples include studies based on trade-off theory (Opler et al., 1999), pecking order theory (Myers & Majluf, 1984), free cash flow theory (Jensen, 1986), financial constraints (Ang & Smedema, 2011; Luo, 2011), corporate governance structure (Kuan et al., 2011; Ozkan & Ozkan, 2004), strategic concern (Fresard, 2010), type of business organization (Duchin, 2010), uncertain preventive demand (Bates et al., 2009), problem of debt agency (Niskanen et al., 2010), and extent to which legal norms protect investors (Dittmar et al., 2003).

⁵ CEO internalization means that the CEO is one of the ultimate controlling owners of the company (Chiang & Lin, 2011). A non-internalized CEO is also called a professional CEO.

analyzes the relationship between two critical demographic variables of the CEO - age and tenure - and corporate cash holdings.

Why is CEO age an important indicator that can be considered for firms' cash holding decisions⁶? This can be illustrated from three perspectives. First, CEO age is an important indicator of an individual's risk attitude, decision-making style, and managerial experience, and thus affects the CEO's cognitive ability, risk tolerance, and emphasis on long- and short-term goals (Andreou et al., 2017; Belenzon et al., 2019; Malm et al., 2021). These attributes have a more direct impact on the firm's cash holding strategy. Second, there is a dynamic and changing relationship between CEO age and management style (Belenzon et al., 2019; Desir et al., 2024), which may change significantly as the CEO grows older, which in turn affects the firm's financial decisions. This dynamic relationship is more direct than the effect of firm age on the firm's financial policy. Finally, CEOs are the most important decision-makers in a firm, and their personal characteristics, especially their age, have a decisive impact on the firm's business strategy and risk management (Chen & Zheng, 2014; Hambrick & Mason, 1984). CEO age reflects their decision-making behavior at different stages of their careers, which directly affects the firm's cash holding policy.

We summarize the contributions of this study below. First, this study differs from previous studies that examine changes in cash holdings from the perspective of firm characteristics or external event shocks (Rao & Tilt, 2016). Instead, we explore the personal characteristics of the key decision maker (CEO) using two important demographic variables: age and tenure. To the best of our knowledge, this is the first study to examine the effects of CEO age and CEO tenure on cash holdings using individual CEO data of Taiwan's listed firms. Under the Personal Data Protection Law in Taiwan, it is not necessary to disclose the age of a CEO. There are no empirical studies examining the effects of CEO age and tenure on corporate cash holdings in Taiwan due to the extreme difficulty of collecting CEO personal information. The main contribution of our study is the collection of CEO age and tenure data through manual methods to make up for the gap in research on this topic.

Second, previous studies on the impact of CEO age and tenure on corporate decision making have mainly focused on developed countries (Ali & Zhang, 2015; Chen et al., 2019; Serfling, 2014; Yim, 2013). As such, there is a lack of evidence from emerging markets. Our findings make up for the gap in this research topic. Third, our collected data show that more than 40% of listed companies in Taiwan (during the period 2010-2019) have internalized CEO⁷, which differs from developed countries in which most CEOs are professional. Thus, we further examine whether internalized CEO affects firm cash holdings and how internalized CEO moderates the effects of CEO characteristics on cash holdings. Fourth, we use a structural equation model to observe how CEO age and tenure affect firm cash holdings. Finally, we observe the CEO tenure in the current company to examine observation period and threshold period effects on firm cash holdings. Our study differs from previous studies in that we analyze CEO individual data instead of firm data (Hu, Blettner, & Bettis, 2011; Intintoli & Kahle, 2016; Simsek, 2004, 2007). Moreover, we examine the effect of CEO internalization on the observation period and threshold period effects.

The rest of this study is organized as follows: Section 2 includes the literature review and hypotheses development. Section 3 introduces the research methods. Section 4 presents the empirical results. Finally, Section 5 is the conclusion.

2. Theories and Empirical Hypotheses

From a risk-taking perspective, younger CEOs tend to take on more risks, but as they reach a certain age threshold, their risk-taking behavior declines. This pattern is particularly evident in family-owned businesses, where near-retirement CEOs may reverse their earlier conservative behavior, leading to an inverted U-shaped relationship between CEO age and risk-taking (Yeoh & Hooy, 2022b)⁸.

Serfling (2014) points out that older CEOs typically adopt less risky investment policies. They tend to invest less in research and development, prefer more diversified acquisitions, and maintain lower operational leverage. This conservative approach reduces company risk and aligns with the preferences of aging CEOs. Liu et al. (2016) find that younger CEOs, aiming to increase their visibility, tend to take on more risk.

Regarding the relationship between CEO tenure and risk factors, as CEO tenure increases, their skills become more specialized in the company's field, reducing their inclination to engage in risky activities (Leng & Pan, 2023). Çolak and Korkeamäki (2021) note that shorter-tenured and more mobile CEOs might adopt higher-risk policies, while Tarus (2020) find that long CEO tenures negatively impact risk management. However, a strong governance structure can mitigate the negative effects of CEO tenure on risk management.

As the human and financial capital of CEOs become tied to the company, CEO risk tolerance is closely correlated

⁶ We thank the referee for this suggestion.

⁷ Based on 7,249 observations, about 80% of listed firms have ultimate controlling shareholders. Moreover, 5,201 observations are of family firms and 3,445 observations are of firms with internalized CEO, accounting for 47.52% of the total sample.

⁸ We thank the referee for providing this comment.

with the riskiness of corporate policies. In general, risk-averse CEOs reduce the firm's risk by adjusting cash holdings to a level that aligns with their risk attitude (Almeida et al., 2004). In other words, CEO risk attitude is reflected in the firm's risk-taking behavior (Cain & McKeon, 2016) and cash holdings (Feng & Rao, 2018). Based on this link, researchers have developed several theories on the relationship between CEO age or tenure and a firm's cash holdings.

As the human capital and financial capital of CEOs are bound to the company, CEO risk tolerance is highly correlated with the riskiness of corporate policies. In general, risk averse CEOs reduce the firm's risk by adjusting cash holdings to a level that corresponds to their risk attitude (Almeida et al., 2004). In other words, CEO risk attitude is reflected in the firm's risk-taking behavior (Cain & McKeon, 2016)⁹ and cash holdings (Feng & Rao, 2018)¹⁰. Based on this link, researchers have developed several theories on the relationship between CEO age or tenure and cash holdings of firms.

2.1 CEO Age and Corporate Cash Holdings

There are several essential theories regarding the effects of CEO age on corporate cash holdings, including career concerns, signal hypothesis, and resource dependency hypothesis. CEO career concerns are related to future compensation and career development (Xie, 2015). Younger CEOs may be more concerned about their career prospects than older CEOs due to the expectation of more working years. Thus, younger CEOs are less likely to invest in high-risk projects, as they may reflect poorly on them as managers if they do not perform well (Hong et al., 2000). Under the career concern hypothesis, younger (older) CEOs are less likely to invest in high-risk projects based on the preventive motive of reputation. Thus, firms with younger CEO hold more (less) cash.

Under the condition of asymmetric information, the signal hypothesis assumes that young CEOs do not yet have a reputation as savvy managers. They invest aggressively and take greater risks to demonstrate their competence and gain recognition (Serfling, 2014). Younger CEOs have a better ability to learn and tend to possess modern technological knowledge and innovative thinking, in addition to sufficient confidence in their decisions (Barker & Mueller, 2002) and active engagement in risky investments (Prendergast & Stole, 1996). An aggressive investment style increases the need for investment capital, leading to lower levels of cash holdings (Lin et al., 2011). In contrast, older CEOs have more work experience and are less likely to pursue risky investments to signal superior capabilities. Thus, under the signal hypothesis, CEO age is positively related to firm cash holdings (Orens & Reheul, 2013).

The resource dependency hypothesis indicates that an increase in CEO social capital¹¹ causes a decrease in the precautionary motive for firm cash holdings. Cash management focuses on the holding of cash as a precaution, flexibility of cash, and minimization of opportunity costs. With increasing age, CEOs accumulate extensive experience and more robust social networks, political and business group connections, and social resources. The marginal effect of CEO social capital on increased corporate risk-taking should be more potent when investor protection standards are weak and financial markets are underdeveloped (Ferris et al., 2019). Based on the resource dependency hypothesis, CEO age negatively affects firm cash holdings.

From a theoretical standpoint, the career concerns hypothesis, signaling hypothesis, and resource dependence hypothesis offer divergent views on the relationship between CEO age and cash holdings, previous empirical results regarding the relationship between CEO age and firm cash holdings are mixed too. Orens and Reheul (2013) investigated the impact of CEO age on firm cash holdings within a cohort of privately held Belgian SMEs, uncovering a tendency among older CEOs to prioritize the precautionary motives for holding cash over opportunity costs. Consequently, firms led by older CEOs exhibited higher cash reserves compared to those with younger CEOs. Gaio et al. (2022) explored the determinants of excess cash holdings among startups in Portugal, revealing a predilection for older founders to maintain greater cash reserves. Conversely, Guha and Rahim (2019) found that older CEOs hold less cash, suggesting a shift in motivation towards establishing a lasting legacy as CEOs age. Supporting the notion of a negative correlation between CEO age and cash holdings. Winarta et al. (2018) corroborated this trend through their research on 24 firms listed on the LQ45 index. This implies that the relationship between CEO age and cash holdings may be moderated by other variables, such as the findings of Yeoh and Hooy (2022a), which indicated a positive correlation between CEO age and corporate cash reserves. Additionally, this relationship was found to be moderated by the role of legal origin, with the positive effect diminishing in countries with better investor protection and financial development. In jurisdictions governed by common law, German law, and in post-socialist countries, older CEOs were found to hold less cash. Thus, understanding the relationship between CEO age and cash holdings necessitates consideration of firm-specific or national characteristics. The governance structure of publicly listed companies in Taiwan is predominantly characterized by family control, with a preference for family members to assume the CEO role. Family CEOs are subject

⁹ Cain and McKeon (2016) examine the effect of CEO personal risk on firm risk using delta and vega as proxies for CEO personal risk. The riskiness of a firm's projects is measured as the annualized standard deviation of monthly stock returns.

¹⁰ Feng and Rao (2018) use delta and vega as proxies of CEO personal risk, to assess the impact of CEO personal risk on cash holdings.

¹¹ Following Woolcock (1998), we define social capital as information, trust, and norms of reciprocity inherent in a social network.

to relatively less performance pressure, thereby diminishing their motivation to signal managerial capabilities to the market. Furthermore, Taiwanese family businesses are largely structured as conglomerates and maintain favorable relationships with the government, possessing extensive networks within the business community and access to a plethora of external resources. Given these attributes, it is anticipated that the negative relationship between CEO age and cash holdings will outweigh any positive correlation. Therefore, we have Hypothesis 1A, which is stated below:

Hypothesis 1A: Older (younger) CEOs manage companies with fewer (more) cash holdings.

Yeoh and Hooy (2022b) find that CEO age has an inverse U-shaped relationship with CEO risk attitude, which is linked to company cash holdings (Feng & Rao, 2018; Tong, 2010). Therefore, we also explore cash holdings of firms as CEOs age. Our Hypothesis 1B is stated below:

Hypothesis 1B: There are smaller decreases in the company's cash holdings as CEO ages.

2.2 CEO Tenure and Corporate Cash Holdings

CEO tenure may alter CEO risk attitudes, including experience accumulation (Chen & Zheng, 2014; Lim & Lee, 2019; Simsek, 2007), power enhancement (Chen & Zheng, 2014), and human capital appreciation (Chen & Zheng, 2014). In this section, we discuss the impact of CEO tenure on firm cash holdings from three perspectives: work experience, CEO power, and human capital.

The longer a CEO's tenure, the more experience he/she accumulates, and the more his/her managerial skills are enhanced (Simsek, 2007). As such, he/she is better able to identify favorable investment opportunities (Allen, 1981; Simsek, 2007), reducing the probability of making mistakes while increasing risky investments (Simsek, 2007), leading to fewer cash holdings.

The accumulated work experience of a CEO may result in overconfidence (Bernardo & Welch, 2001; Chen, Ho, & Yeh, 2020), which is related to past and future decisions. The effects of these two types of overconfidence on cash holdings differ. A CEO who has been overconfident in the past is convinced that his/her existing management style applies to the future and, thus, prefers to maintain the status quo (McClelland et al., 2010). The company's cash holdings increase to prevent uncertain capital needs.

In contrast, if a CEO's overconfidence is due to experience in the past and reflected in future investment projects, it may lead to an overestimation of the return on investment projects or an underestimation of the risk of investment projects (Landier & Thesmar, 2008), which in turn leads to overinvestment (Heaton, 2002) and decrease in cash holdings.

The longer a CEO's tenure, the greater his/her understanding of the company culture and operating mechanisms and the more internal and external information and resources he/she controls. CEO tenure leads to significant influence and control over the company (Allen, 1981), with increases in power and status (DeBoskey et al., 2019). The longer a CEO's tenure, the deeper the relationship between the CEO and the board and the more likely the CEO is to influence the board, resulting in a less independent board (Hermalin & Weisbach, 1998) and reduced internal oversight (Hambrick & D'Aveni, 1992). Less supervision of the CEO leads to increased benefits of control and increased cash holdings for conduct management defense (Hermalin & Weisbach, 1998). A CEO's interest in continuing control reduces risk-taking (Bertrand & Mullainathan, 2003). Thus, more powerful CEOs prefer to have more cash holdings.

As CEO tenure increases, investment in and value of human capital increase (Chen & Zheng, 2014) and the cost of job transition also increases. Based on the trade-off theory, CEOs emphasize consolidation of their position. Thus, in addition to a more conservative risk attitude (Chakraborty et al., 2007), a stronger managerial defense motive leads to holding of more cash to control expansion of capital.

Previous studies show mixed results regarding the relationship between CEO tenure and corporate cash holdings. Lim and Lee (2019) find that longer CEO tenure among Korean listed companies is associated with lower excess cash holdings but shorter CEO tenure only marginally impacts on excess cash holdings¹². Orens and Reheul (2013) also find that longer CEO tenure is associated with higher cash holdings among private SMEs in Belgium. In contrast, Suherman et al. (2021) investigate the relationship between CEO tenure and corporate cash holdings in Indonesia and find that CEO tenure is negatively associated with corporate cash holdings, while Guha and Rahim (2019) demonstrate no association between CEO tenure and cash holdings.

Overall, the relationship between CEO tenure and company cash holdings is determined by the total effects of work experience, CEO power, and human capital. Taiwan's corporate governance environment is still immature and, thus, we expect that the impacts of CEO power and human capital are greater than that of experience and there is a positive relationship between CEO tenure and corporate cash holdings. Hypothesis 2A is stated below:

¹² The empirical results of a study by Lim and Lee (2019) show that the estimated coefficient of CEO tenure is negative, the estimated coefficient of the square term of CEO tenure is positive, and the relationship between CEO tenure and cash holdings is U-shaped. On regression, tenure's impact on cash holdings changes from negative to positive within 2 years.

Hypothesis 2A: As a CEO's tenure increases, a firm's cash holdings increase.

We further analyze whether the CEO's risk attitude changes to differing degrees as the CEO's tenure increases, resulting in other effects of CEO tenure on cash holdings and whether there is a non-linear relationship between CEO tenure and cash holdings. Our Hypothesis 2B is stated below:

Hypothesis 2B: As CEO tenure increases, company cash holdings level is associated with CEO tenure.

2.3 Moderating Effects of CEO Internalization

Internalized CEOs are subject to less supervision by the board of directors than non-internalized CEOs. This mitigates the pressure from external monitoring. With less information asymmetry with shareholders and fewer worries about their career path, CEOs can pay more attention to the interests of controlling shareholders than those of ordinary shareholders and have greater power over and more influence on the company's decisions (Zona, 2016). CEO internalization is distinct from governance mechanisms in which managers and owners are separated, so traditional agency theory cannot explain the behavior of internalized CEOs. Based on the above differences, internalized and non-internalized CEOs exhibit different cognitive frames, views, values, and expectations, resulting in different effects on firm cash holdings.

Compared with non-internalized CEOs, internalized CEOs are less concerned about their careers because they are less likely to be fired for poor performance (Tsai et al., 2006). Therefore, internalized CEO age has a less negative effect on cash holdings from the perspective of career concern. In addition, the internalization of the CEO reduces the incentive to signal managerial ability (Michiels et al., 2013). We expect that the signal effect of internalized CEO age on cash holdings is smaller than that of non-internalized CEO age.

From equations 3 and 6 in Table 4, the estimated coefficient of CEO age is significantly negative and the estimated coefficient of the cross product between CEO age and internalized CEO is significantly positive, but smaller than the estimated coefficient of CEO age. This means that internalized CEO can moderate the negative impact of CEO age on cash holdings.

From the above discussion, we expect that CEO internalization positively moderates the relationship between CEO age and cash holdings. In other words, CEO internalization mitigates the negative effect of CEO age on cash holdings. From this, we have Hypothesis 3, which is stated below:

Hypothesis 3: CEO internalization mitigates the negative impact of age on firm cash holdings.

CEOs with longer tenure accumulate more experience. As such, they are less likely to make mistakes and more likely to take on high-risk projects, leading to decreased cash holdings (Tsai et al., 2006). Internalized CEOs tend to focus on maintaining controlling shareholders' assets more than non-internalized CEOs. Relative to non-internalized CEOs, internalized CEOs with experience are associated with a smaller increase in risk-taking and a smaller decrease in cash holdings.

Internalized CEOs have more power than non-internalized CEOs and are subject to less monitoring, so more powerful internalized CEOs hold more cash for their own self-interest (Liu et al., 2014). In addition, from the perspective of human capital, internalized CEOs are less likely to be fired, so they do not need to control capital expansion by increasing cash holdings to consolidate their position.

The relationship between CEO internalization and the effect of CEO tenure on cash holdings is determined by the total effects of work experience, CEO power, and human capital. Most of the effects discussed above support a negative relationship. Thus, we expect longer tenure of internalized CEO to lead to less of an increase in cash holdings than that of non-internalized CEO. Hypothesis 4 is stated below:

Hypothesis 4: CEO internalization decreases the positive effect of CEO tenure on firm cash holdings.

2.4 Observation Period Effect of CEO Tenure

Hu, Blettner, and Bettis (2011) propose a three-reference-point risk-taking model and suggest that decision makers are more likely to be risk averse if approaching the "success point" and risk seeking if past the "success point." The observation period for new CEOs and the industry average CEO tenure are the two reference points for CEO tenure. We decomposed these into pure tenure effect, observation period effect, and threshold effect.

Under asymmetric information, the board of directors is uncertain about the new CEO's ability. Therefore, stricter monitoring is implemented during the initial observation period of the CEO appointment to confirm whether he/she meets the board's expectations and his/her appointment should continue after the observation period.

In general, the observation period is defined as a CEO's first two years in office (Intintoli & Kahle, 2016). Thus, to continue his/her appointment, a new CEO tries to achieve the goals set by the board as soon as possible (Westphal & Fredrickson, 2001). Quigley and Hambrick (2012) point out that a new CEO temporarily conceals his/her personal risk appetite and adopts a prudent approach during the observation period. New CEOs tend to hold more cash to cope with

future uncertainty in the external environment and reduce losses from insufficient capital (Bates et al., 2009; Duchin, 2010; Fresard, 2010). Thus, we expect that the observation period effect leads to increased cash holdings. However, if the CEO is internalized, because there is less observation period pressure, the observation period has less effect on cash holdings. Hypothesis 5A and Hypothesis 5B are stated below:

Hypothesis 5A: The observation period effect increases firm cash holdings.

Hypothesis 5B: CEO internalization decreases the positive effect of the observation period on firm cash holdings.

2.5 Threshold Effect of CEO Tenure

To explore the tenure effects on firm cash holding policies, in addition to the observation period effect discussed above, the threshold period of CEO tenure was defined. Since there is asymmetric information regarding CEO capabilities in the CEO labor market, CEO tenure is often used as a signal to evaluate a CEO's capabilities (Bergh, 2001). If a CEO's tenure exceeds the industry-average (tenure threshold), this is an indicator that his/her management ability exceeds that of most CEOs in the same industry. This might influence his/her reputation and the opportunity for career development or salary bargaining in the future.

Hu, Kale, et al. (2011) suggest that if there is a tenure threshold effect, when a CEO's tenure is close to the industry-average, he/she is motivated to consolidate his/her position to extend his/her tenure. To reduce loss due to risk from insufficient capital during changes in the external environment, it is expected that a CEO's desire to exceed the target tenure threshold leads to an increase in the firm's cash holdings as the CEO approaches the target tenure threshold, which is called the threshold effect. In contrast, internalized CEOs face less pressure than non-internalized CEOs and, thus, have less entrenchment motivation as their tenure approaches the industry-average. Thus, we have Hypothesis 6A and Hypothesis 6B as stated below:

Hypothesis 6A: The CEO's target tenure threshold increases the firm's cash holdings.

Hypothesis 6B: CEO internalization decreases the positive effect of the target tenure threshold on firm cash holdings.

3. Empirical Design and Variable Descriptions

3.1 Data

In this study, we observe listed firms in the non-financial industry in Taiwan from 2010 to 2019¹³. This study manually collected CEO age and tenure data, focusing on the period from 2010 to 2019 for three main reasons¹⁴. First, during this 10-year period, Taiwan's economic environment was relatively stable, making the research findings more representative of the impact of CEO decisions on corporate cash holdings. Second, a 10-year sample period is sufficiently long to observe the long-term effects of CEO age and tenure on cash holdings. Finally, since Taiwan began implementing corporate governance reforms in 2002, listed companies have been required to make continuous improvements. By 2010, the implementation of the audit committee system stabilized corporate governance policies (Chuang et al., 2023). This timeframe selection helps eliminate external interference from frequent policy changes, ensuring that the research results are more accurate and reliable.

The industry classification of each company is based on the Standard Industrial Classification Code (SIC Code) assigned by the Taiwan Stock Exchange. Our data for all variables is mainly from the Taiwan Economic Journal (TEJ) and the Market Observed Post System (MOPS) of the Taiwan Stock Exchange. Most importantly, the age of the CEO and the dates of CEO election and inauguration are collected manually since disclosure of the birthdate or age of directors and executives of listed companies is not mandatory under Taiwan law¹⁵.

As data on CEO age are collected from different sources, we must ensure that there are no inconsistencies¹⁶. The annual list of directors and supervisors, the date of the initial appointment of managers, the date of election, the list of the last five CEOs of each company, and the date of inauguration are from the corporate governance database of TEJ and announcements of CEO changes in MOPS. We also confirm CEO birth year using Wikipedia, print media articles, and Google Web Search Engine¹⁷. Our study sample of 7,249 observations with complete variables accounts for 88.74%

¹³ The reason for excluding financial companies (Stock Code 2801-2897 and 5876-6024) is that the financial industry differs from other industries in that a company's operations must not only comply with Company Law, but also with relevant financial regulations. More importantly, financial statements are prepared in a different manner than in other industries.

¹⁴ We thank the referee for providing this comment.

¹⁵ TEJ provides basic company information, financial statements, shareholder structure, board of directors' structure, list of directors and officers, and their education. Information about changes in the CEO of a company are obtained from MOPS.

¹⁶ Since data are obtained from various channels, to solve date distortions which can lead to data errors, we cross-reference the information obtained from TEJ with that from MOPS.

¹⁷ CEOs are often recognized as outstanding alumni by their alma maters. After obtaining the name of the school from which

of the total sample of 8,169 observations¹⁸.

Among the total observations, the highest proportion (3,809 observations, 46.63%) is in the electronics industry. We have complete data for 7,249 observations. Among them, the electronics industry accounts for 3,429 or 47.30% of the total sample.

In Taiwan, the government provides more tax reductions and advantages for stock listings in the electronics industry than for any other industry. The electronics industry has the largest number of firms requiring continuous R&D of innovative technologies to maintain competitiveness. The second largest number of listed companies is in the chemical, biotechnology, and medical industry, which also emphasizes R&D. These two industries include more than half of all non-financial companies in Taiwan.

3.2 Measurements of Variables

3.2.1 Dependent variables

The dependent variable is the level of company cash holdings, defined as the sum of cash and cash equivalents divided by the value of the company's total assets and multiplied by 100, as in our empirical models (Chen, Ho, & Yeh, 2020; Ozkan & Ozkan, 2004).

3.2.2 Independent variables

A. CEO age

CEO Age is measured as the data year minus the birth year. If there is a change of CEO in the observed year, the CEO age variable is the weighted average age of the CEO with the proportion of time spent as the CEO of the company in that year as the weight.

B. CEO tenure

CEO Tenure refers to the total number of years of service as CEO of the company, measured in years from the date of appointment to the end of the current year¹⁹. If there is a change in CEO, the number of CEOs in that year exceeds one and the CEO tenure of the company in that year is measured by the same method as CEO age. In other words, CEO tenure is calculated as a weighted average.

C. Observation period

The *Observation Period* is represented by a dummy variable used to determine whether the CEO's tenure falls within the probationary period of a new appointment. Following the method of Intintoli and Kahle (2016), if the CEO's tenure at the end of the year is less than two years, the observation period variable is set to 1; otherwise, it is set to 0.

D. Threshold period

The *Threshold Period* is represented by a dummy variable used to determine whether the CEO's tenure is approaching the target tenure (industry average CEO tenure). If the difference between the target tenure and the CEO's current tenure is between 0 and 2 years, the threshold period variable is set to 1; otherwise, it is set to 0.

E. Control Variables

On the basis of the existing empirical literature, we include other factors that affect cash holdings as control variables: financial constraints (Almeida et al., 2004; Mun et al., 2020), leverage (Mun et al., 2020; Opler et al., 1999), market-to-book ratio (Chen, Yang, et al., 2020; Kim et al., 1998), net working capital (Mun et al., 2020), capital expenditures (Chen, Yang, et al., 2020; Mun et al., 2020; Rizwan & Javed, 2011), R&D expenditures (Chen, Yang, et al., 2020; Lim & Lee, 2019), and diversification (Serfling, 2014; Subramaniam et al., 2011).

The literature on financial constraints encompasses a multitude of measurement approaches. Chen and Wang (2012) employ a combination of metrics including the KZ index, cash-cash flow sensitivity, investment-cash flow sensitivity, and the WW index to gauge financial constraints, with both the KZ and WW indexes comprising composite indices

a CEO graduated, through the database of directors and managers of TEJ, we find his/her graduation year based on the list of outstanding alumni of that school and from that deduce his/her year of birth. Other online channels include the Central Research Institute's Modern People Information Integration System and the Taomingren website (www.taomingren.com). Publications referenced by written data inquiry include the Taiwan Business and Industry Association's *The People's Journal of the Republic of China* and the Central News Agency's *Who's Who in Taiwan*, as well as *Contemporary Celebrities of the Republic of China* published by the Chung Hwa Book Company and *Who's Who in Taiwan* published by Feng Yun Forum Press.

¹⁸ Some CEOs refuse to provide their birth year to protect their personal data.

¹⁹ In our study, there are a total of 195 reappointed CEOs, including 186 who reached tenure once and 9 who reached tenure twice. Among them, 18 did not reach tenure during the study period (2010-2019). Individual CEO Tenure is measured as the total cumulative number of years as CEO of the company as of the Observed year.

formed by multiple factors. In addition, firm characteristics serve as proxy variables, such as credit ratings (Bottazzi et al., 2014) and firm size (Almeida et al., 2004), given that the outcomes of credit ratings signify the likelihood of a firm's default and, by extension, the ease or difficulty with which it can secure funding (Bottazzi et al., 2014). Given the heightened financing challenges faced by smaller firms, firm size is utilized as a proxy for financial constraints (Almeida et al., 2004). The academic community remains divided over which measures most effectively capture financial constraints (Farre-Mensa & Ljungqvist, 2016). This study opts to follow the approach of Almeida et al. (2004) in using firm size to assess dummy variables financial constraints, financial constraints = 1 if the firm size is lower than the industrial average firm size; otherwise, it is = 0.

When the independent variables in a regression model are highly correlated, it results in multicollinearity, which can lead to biased estimations. To ascertain the absence of multicollinearity in the model, the Variance Inflation Factor (VIF) for each variable within the models is calculated. A VIF value exceeding 10 indicates a significant multicollinearity issue within the model. This study provides the maximum VIF value for all variables in each model, revealing a highest VIF of 4.479 across all models, which is below the critical threshold of 10. This suggests that none of the models suffer from severe multicollinearity issues. The definitions and measurements of variables are described in Table 1.

Table 1
Definitions and Measurements of Variables

	Variables	Definitions and Measurements
Dependent Variable	Cash Holdings	We define firm cash holdings as the ratio of cash and equivalent cash/total assets) $\times 100$.
CEO Characteristic Variables	CEO Age	We use the data year and birth year to measure the CEO's age in the current year. There might be multiple CEOs for one firm in the current year due to CEO changes; the proxy variable of CEO age is the weighted average age calculated based on the proportion of each CEO's tenure in the current year.
	CEO Tenure	The number of years the CEO has been in his/her position until the current year. If the CEO is returning, the previous tenure period is included. If there is a CEO change in the current year, the weighted average tenure is calculated as the proxy variable of the CEO tenure based on the weight, which is the proportion of the tenure of each CEO in the current year.
Control Variables	Financial Constraints	Following Almeida et al. (2004), firm size is used to distinguish whether financial constraints exist. The natural logarithm of total assets measures firm size. Then, the industrial average firm size for each year is calculated. We define a dummy variable, financial constraints = 1 if the firm size is lower than the industrial average firm size; otherwise, it is = 0.
	Leverage	Leverage represents the debt ratio, measured as (total liabilities/total assets).
	Market to Book	Market to Book represents the market-to-book value ratio, measured as (the market value of equity/book value of equity).
	Net Working Capital	Net Working Capital represents non-cash net working capital, measured as ((current assets - current liabilities - cash)/total assets).
	Capital Expenditures	Capital expenditures are measured as (expenditures for purchases of fixed assets, intangible assets, and other long-term assets/total assets).
	R&D Expenditures	The R&D expenditure ratio is measured as (R&D expenditures/total assets).
	Diversification	Diversification represents the degree of diversification of the company, measured by the number of departments in the company.
Instrumental Variables	Life Expectancy	We use the annual average life expectancy for each CEO at 30 years of age, as the instrumental variable for CEO age. In order to include all the Observed in the sample, the average remaining life expectancy of those born in the year that the CEO turns 30 is used since the average remaining life expectancy of the national birth year is calculated from 1951 and the year of birth of the sample CEOs ranges from 1923 to 1987.

	Variables	Definitions and Measurements
	Industry-average CEO Tenure (-1)	Measured by the average CEO tenure of all companies in the industry in the previous year, this serves as an instrumental variable for CEO tenure.
Tenure Effect	Observation Period	Represents the dummy variable of whether a CEO is a new appointment with tenure within the observation period. Following Intintoli and Kahle (2016), we set the observation period = 1 if the CEO's tenure as of the end of the year is less than two years; otherwise, it is = 0.
	Threshold Period	Represents the dummy variable of whether the tenure of a CEO is in the tenure threshold period. We set threshold period = 1 if the target tenure (industry average) minus the CEO's current tenure is between 0 and 2; otherwise, it is = 0.

3.3 Empirical Design

3.3.1 Panel regression and model diagnosis

This study is based on all listed companies in the non-financial sector in Taiwan from 2010 to 2019. Since the study variables contain cross-sectional and time-series characteristics, panel regression is used for empirical analysis to match the characteristics of the study data. The choice of model is based on the Hausman test. If the results are significant, the fixed effect model is used for estimation. If the results are not significant, the random effect model is used.

To ensure the consistency, unbiasedness, and validity of the model estimation coefficients, co-linearity, heterogeneous variation, and self-correlation are diagnosed and the obtained estimation results are the Best Linear Unbiased Estimate (BLUE). We use $VIF < 3$ to define no co-linearity problem. Thus, our models are modified by deleting the variables with lower explanatory power from among the variables causing the co-linearity problem.

Another problem is that of collinearity between CEO age and CEO tenure. Both variables are time dependent. If a CEO is continuously employed during the study period, his/her tenure and age increase by 1 each year. The deletion of variables is, thus, not appropriate. Fortunately, the correlation coefficient between the age and tenure of CEOs in the sample is only 0.28, which implies that this is not a significant problem.

The most common method for examining moderate variables is to include the interaction terms between the explanatory and moderate variables in the regression model. However, the interaction terms are easily collinear with the explanatory or moderate variables. We follow the approach of Villa et al. (2003) to first calculate the standardized scores (Z scores) for the explanatory and moderate variables, then conduct the interaction terms by the standardized scores rather than the original values.

If the residuals have heterogeneous variation, the least square estimate is still an unbiased estimate, but not BLUE. The cross-sectional and time-series residuals are examined for heterogeneous variation and the cross-sectional or period-weighted least square estimation method is chosen to correct the model based on the results to ensure that the results of the diagnosed and corrected empirical model are realistic. The Durbin Watson method is used to check the self-correlation of the residuals. The Durbin Watson values of all the regression models in this study are around 2 after using the weighted least square method to correct for the heterogeneous variation of the residuals, which means that there is no self-correlation problem in the residuals of the model after correction.

3.3.2 Endogeneity problem

Serfling (2014) points out that CEO age is related to the firm's investment decisions and risk, implying that CEO age is an endogenous variable. CEO tenure is also related to the board of directors' decision to reappoint the CEO, and the board's main determinant of whether to reappoint the CEO is the CEO's performance. Thus, CEO tenure might also be an endogenous variable.

To address the endogeneity of CEO age and tenure, we use a two-stage panel regression model (2SLS) in which we choose the instrumental variables for running the regression of CEO age or/and tenure in the first stage. Cline and Yore (2016) and Serfling (2014) use the consumer price index (CPI) of the year of birth of the CEO as the instrumental variable for CEO age. However, the earliest available CPI data in Taiwan is from 1981, while the birth years of the CEOs in our study range from 1923 to 1987. Thus, it is inappropriate to use CPI as an instrumental variable of age in this study. We thus use the national average life expectancy for the year each CEO turned 30 as the instrumental variable for CEO age²⁰. In addition, the year-industry average tenure of CEOs in the previous year is used as the instrumental

²⁰ The data on national average life expectancy are published by the Census and Statistics Department of the Ministry of Health and Welfare of Taiwan. To examine national average life expectancy by tracking CEO age, we choose the age of 30 to cover most of the CEO dataset.

variable for CEO tenure, following Chen et al. (2019) and Khan et al. (2020).

4. Empirical Results

4.1 Descriptive Statistics

The descriptive statistics of the full sample are presented in Panel A of Table 2. Mean (median) cash holdings are 16.160 (13.286). The lowest cash holdings level is less than 1%, while the highest cash holdings level is 93.226% of the total assets. Overall, one fourth of observed cash holdings are below 7% of the total assets.

The youngest CEO is 23 and the oldest CEO is 92, both of whom are family CEOs. Very young and very old CEOs are very unlikely in companies governed by professional managers but, rather, in companies of a special control type such as family companies.

The average tenure of CEOs is 9.933 years, and the longest tenure is more than 50 years. Most CEOs with extremely long tenure work in family companies. Most CEOs who step down after less than one year in office are from companies governed by professional managers. Out of 180 cases in this study sample, only 54 were of family companies and among those only 16 CEOs were family members. This shows that there are significant differences in the selection and appointment of CEOs, as well as their supervision and evaluation.

The lowest total company assets are 1.727 (millions). Hon Hai possesses the highest total company assets, reaching 340,721 (millions) in 2018, which is about 200,000 times that of the smallest company. R&D expenditures are mainly related to the industry and the company's operating policy. Traditional industries have a lower demand for R&D. However, for the electronics and chemical, biotechnology, and medical industries, R&D is indispensable for enhancing competitiveness and creating company value. The 25% quantile is equal to 0, meaning that at least 1/4 of the observed values are from companies that are not engaged in R&D.

Panel B in Table 2 lists the means and standard deviations of the variables for the CEO age groups. By dividing the sample into three groups of equal number, we find the cut-off points of 55 and 62 years old. CEOs that are older than or equal to 62 are classified as old CEOs. CEOs aged less than or equal to 55 are classified as young CEOs and CEOs aged over 55 to under 62 are classified as middle-aged CEOs. Panel B of Table 2 shows the results of two-by-two comparisons of the main variables for the old and young groups and the middle-aged and young groups. From the mean difference test results, there are significant differences between the old and young groups for all variables. Old CEOs are associated with longer CEO tenure, larger company size, higher level of debt, higher capital expenditures, higher level of diversification, fewer financial constraints, lower market-to-book ratio, lower net working capital, and lower R&D expenditures. For the comparisons of middle-aged and young groups, there are only three variables for which there are no significant differences, market-to-book ratio, net working capital, and R&D expenditures.

Panel C in Table 2 shows the means and standard deviations of the main variables using the same method for grouping CEOs, this time based on tenure. Short CEO tenure is defined as tenure of less than or equal to 3.83 years, medium CEO tenure is defined as tenure of between 3.83 and 10.84 years, and long CEO tenure is defined as tenure of greater than or equal to 10.84 years. The same approach as in Panel B is used to determine whether there are significant differences in the key variables across the CEO tenure subgroups. There are no significant differences in cash holdings, market-to-book ratio, capital expenditures, or R&D expenditures.

Table 2
Descriptive Statistics

Panel A: Full Sample Descriptive Statistics							
	Mean	Median	SD	Min	Max	25th	75th
Dependent Variable							
Cash Holdings	16.160	13.286	12.760	0.133	93.226	6.661	22.267
CEO Characteristic Variables							
CEO Age	59.138	59.000	8.848	23.000	92.000	54.000	64.000
Adjusted CEO Age	0.000	-0.044	8.310	-29.952	31.786	-5.164	5.000
CEO Tenure	9.933	7.003	8.938	0.082	50.030	3.167	13.844
Adjusted CEO Tenure	0.000	-2.646	8.913	-14.311	38.997	-6.525	4.204
Control Variables							
Total Assets (million NTD)	3,448.821	697.943	13,324.802	1.727	340,721.661	309.904	1,753.771
Financial Constraints	0.550	1.000	0.497	0	1	0	1
Leverage	0.431	0.436	0.182	0.000	1.090	0.297	0.556
Market-to-Book Ratio	1.636	1.240	1.971	0.220	75.260	0.840	1.880
Net Working Capital	0.097	0.088	0.176	-0.759	0.920	-0.011	0.203
Capital Expenditures	0.055	0.062	0.290	-2.118	6.181	-0.002	0.132
R&D Expenditures	3.725	1.290	15.485	0.000	835.490	0	3.79
Diversification	4.190	3	3.294	1	54	2	6

Panel A: Full Sample Descriptive Statistics									
	Mean		Median		SD	Min	Max	25th	75th
Instrumental Variables									
Life Expectancy		42.838		42.990	1.439	36.390	48.200	42.220	43.410
Industry average CEO Tenure(-1		9.292		9.198	1.878	3.589	14.560	8.240	10.699
Panel B: Summary Statistics by CEO Age									
	Young CEO ¹		Middle-Aged CEO		Old CEO		Different		
	(A)		(B)		(C)		(C)-(A)	(B)-(A)	
	Mean	SD	Mean	SD	Mean	SD	t Stat.	t Stat.	
Dependent Variable									
Cash Holdings	0.167	0.131	0.169	0.131	0.144	0.107	-6.534	***	0.453
CEO Characteristic Variables									
CEO Tenure	8.505	7.770	11.327	8.612	14.847	11.741	18.180	***	10.832 ***
Control Variables									
Firm Size	15.659	1.341	16.321	1.457	16.510	1.570	17.740	***	14.988 ***
Financial Constraints	0.620	0.486	0.430	0.495	0.380	0.485	-15.272	***	-12.758 ***
Leverage	0.427	0.186	0.438	0.178	0.445	0.171	3.343	***	2.106 **
Market-to-Book Ratio	1.653	2.170	1.681	1.604	1.517	1.247	-2.977	***	0.448
Net Working Capital	0.102	0.180	0.088	0.167	0.084	0.165	-3.470	***	-2.569 ***
Capital Expenditures	0.034	0.074	0.032	0.085	0.038	0.053	2.023	**	-0.579
R&D Expenditures	3.828	18.291	3.973	6.007	3.004	5.491	-2.844	***	0.284
Diversification	4.085	3.068	4.457	4.094	4.398	3.290	3.069	***	3.092 ***
Obs.	2,410		2,368		2,471				
Panel C: Summary Statistics by CEO Tenure									
	Short CEO Tenure ²		Medium CEO Tenure		Long CEO Tenure		Different		
	(A)		(B)		(C)		(C)-(A)	(B)-(A)	
	Mean	SD	Mean	SD	Mean	SD	t Stat.	t Stat.	
Dependent Variable									
Cash Holdings	0.162	0.127	0.164	0.135	0.165	0.119	0.934		0.742
CEO Characteristic Variables									
CEO Age	57.072	9.115	57.951	8.795	61.258	8.295	10.905	***	2.225 **
Control Variables									
Firm Size	15.961	1.587	15.864	1.478	15.889	1.266	-1.790	*	-2.293 **
Financial Constraints	0.540	0.498	0.570	0.496	0.540	0.499	0.600		2.318 **
Leverage	0.437	0.186	0.435	0.183	0.423	0.186	-2.784	***	-0.470
Market-to-Book Ratio	1.676	2.769	1.635	1.682	1.605	2.769	-1.103		-0.665
Net Working Capital	0.086	0.170	0.089	0.171	0.115	0.170	5.897	***	0.722
Capital Expenditures	0.034	0.087	0.036	0.072	0.033	0.087	-0.560		0.839
R&D Expenditures	4.254	24.896	3.651	12.189	3.375	24.896	-1.643		-1.063
Diversification	4.344	3.779	4.132	3.241	4.144	3.779	-2.063	**	-2.140 **
Obs.	2,462		2,390		2,397				

Note. CEOs aged 55 or under are young CEOs, those aged over 55 but under 62 are middle-aged CEOs, and those aged 62 or over are old CEOs. CEO tenures of less than or equal to 3.83 years are short tenures, those of more than 3.83 and less than 10.84 years are medium tenures, and those of 10.84 years or more are long tenures.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In constructing the empirical model, it is important that the explanatory variables be independent, to avoid collinearity problems. Therefore, the correlation coefficients of the studied variables are used to determine whether there is strong correlation among the explanatory variables. The maximum Pearson correlation coefficient among all the

explanatory variables is 0.29, meaning that there is no strong correlation between the explanatory variables²¹.

4.2 Empirical Results

4.2.1 CEO age and tenure

In Table 3, we first focus on the effect of CEO age in Panel A, then on the effect of CEO tenure in Panel B, and finally on the effects of both CEO age and tenure in Panel C on company cash holdings by considering other controlling variables. We also present panel regression results for Models 1-3 and panel 2SLS regression results for Models 4-5.

In Panel A of Table 3, we consider two CEO age groups in our models, including dummy variable and CEO age. Panel regression of Model 1 shows that compared with young CEOs (aged 55 or below), only old CEOs (aged 62 or over) are associated with significantly lower cash holdings. In Model 2, CEO age has a significant and negative impact on cash holdings. On the other hand, CEO age has a positive relationship at a decreasing rate (non-linear) with cash holdings in Model 3. Interestingly, the estimated coefficients of CEO Age and (CEO Age)² are 0.052 and -0.003 in Model 3. The results of Model 3 differ from those of Model 2, for which the estimated coefficient of CEO age, -0.020, is significant and negative. However, in Model 3 the effect of CEO age on cash holdings changes from positive to negative when the estimated coefficient of CEO age exceeds 8.67. The minimum CEO age in this study sample is 23 and there is only a negative relationship between CEO age and cash holdings for the whole sample. This implies that the empirical results of both Model 2 and Model 3, for the effect of CEO age on cash holdings, are negative and non-linear effect does not exist.

Panel 2SLS regression results in Models 4-5 are consistent with panel regression results in Models 2-3. Summarizing the above empirical results, there are significant negative effects of CEO age and squared term of CEO age on firm cash holdings, which supports only Hypotheses 1A but not Hypotheses 1B. Our findings are consistent with those of Guha and Rahim (2019) for US listed companies. However, they differ from the findings of Belgian SMEs by Orens and Reheul (2013) and Portuguese startups by Gaio et al. (2022).

In Panel B of Table 3, Model 1 results indicate that compared with short CEO tenure, only long CEO tenure has significant and positive association with cash holdings. Models 2-3 show that the longer the CEO tenure, the higher the level of firm cash holdings, with no non-linear effect. The estimation results of the panel 2SLS regression method for Models 4-5 are the same as those of the panel regression method. Both regression models support that CEO tenure has a positive effect on the level of cash holdings, consistent with a study by Orens and Reheul (2013). Our results indicate that there is no non-linear relationship between CEO tenure and cash holdings, which is inconsistent with Lim and Lee (2019). Hypothesis 2A, but not Hypothesis 2B, is supported. These results differ from those of previous studies. For example, Suherman et al. (2021) find negative correlation between CEO tenure and cash holdings, while Guha and Rahim (2019) find no relationship.

Finally, we consider both CEO age and CEO tenure in the same models in Panel C. Results are consistent with those of Panel A and Panel B. Our empirical results provide strong support for hypotheses 1A and 2A.

Table 3
Impacts of CEO Age and Tenure on Corporate Cash Holdings

Panel A: Impact of CEO Age on Corporate Cash Holdings					
Model	Panel Regression			Panel 2SLS Regression	
	Dependent variable = cash holdings			Dependent variable = cash holdings	
	1	2	3	4	5
Constant	4.717 *** (10.101)	6.865 *** (9.192)	4.260 * (1.744)	8.205 *** (9.569)	3.579 (0.228)
Cash Holdings (-1)	0.828 *** (95.453)	0.835 *** (103.815)	0.834 *** (103.531)	0.818 *** (94.471)	0.363 *** (21.164)
CEO Characteristics					
Middle-Aged CEO	0.015 (0.058)				
Old CEO	-0.475 ** (-2.122)				
CEO Age		-0.020 * (-1.955)	0.052 * (1.873)	-0.032 *** (-2.663)	0.004 * (1.909)
(CEO Age) ²			-0.003 **		-0.008 *

²¹ For the sake of rigor, the VIF values of the explanatory variables are examined at the time of implementation of the regression model and a more stringent criterion of VIF < 3 is applied to determine if there is a common linearity problem that could cause estimation bias.

			(-2.120)		(-1.864)
Control Variables					
Financial Constraints	0.617 *** (3.015)	0.409 ** (2.154)	0.416 ** (2.191)	0.340 * (1.728)	0.175 (0.301)
Leverage	-3.313 *** (-5.348)	-5.895 *** (-10.361)	-5.900 *** (-10.368)	-6.486 *** (-10.777)	-10.976 *** (-9.121)
Market-to-Book Ratio	0.091 (1.397)	0.118 ** (2.026)	0.117 ** (2.004)	0.118 * (1.764)	0.282 *** (2.625)
Net Working Capital	-5.750 *** (-9.238)	-5.780 *** (-10.032)	-5.779 *** (-10.029)	-6.593 *** (-10.828)	-24.295 *** (-18.849)
Capital Expenditures	-2.127 * (-1.636)	-0.629 (-0.536)	-0.624 (-0.532)	-4.837 *** (-2.985)	-7.731 *** (-3.620)
R&D Expenditures	0.023 (1.510)	0.061 *** (4.479)	0.060 *** (4.381)	0.119 *** (6.739)	-0.199 *** (-4.498)
Diversification	-0.077 *** (-2.855)	-0.077 *** (-3.086)	-0.077 *** (-3.094)	-0.082 *** (-3.201)	-0.113 ** (-2.025)
Industry Effect	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Random	Random
Adjusted R-squared	0.776	0.769	0.812	0.768	0.830
Max VIF	1.404	1.406	1.406	1.406	1.406
Durbin Watson Statistic	2.121	2.082	2.083	1.989	1.867

Panel B: Impact of CEO Tenure on Corporate Cash Holdings

Obs. = 7,249	Panel Regression		Panel 2SLS Regression		
	Dependent variable = cash holdings		Dependent variable = cash holdings		
Model	1	2	3	4	5
Constant	4.654 *** (13.567)	4.701 *** (15.516)	11.165 *** (37.42)	5.005 *** (13.708)	4.877 *** (12.784)
Cash Holdings (-1)	0.816 *** (130.854)	0.816 *** (145.332)	0.442 *** (46.795)	0.809 *** (120.987)	0.809 *** (120.968)
CEO Characteristics					
Medium CEO Tenure	0.119 (0.665)				
Long CEO Tenure	0.420 *** (2.354)				
CEO Tenure		0.014 * (1.884)	0.037 ** (2.262)	0.019 ** (2.227)	0.046 * (1.855)
(CEO Tenure) ²			-0.001 (-1576)		-0.001 (-1.164)
Control Variables					
Financial Constraints	0.611 *** (3.958)	0.602 *** (4.334)	0.519 *** (2.524)	0.517 *** (3.179)	0.525 *** (3.225)
Leverage	-3.451 *** (-7.389)	-3.452 *** (-8.207)	-2.566 *** (-7.074)	-3.938 *** (-7.739)	-3.944 *** (-7.751)
Market-to-Book Ratio	0.165 *** (4.432)	0.165 *** (4.925)	0.168 *** (5.461)	0.165 *** (4.288)	0.165 *** (4.288)
Net Working Capital	-5.630 *** (-12.452)	-5.609 *** (-13.777)	-10.643 *** (-21.141)	-5.914 *** (-12.287)	-5.923 *** (-12.305)
Capital Expenditures	-0.800 *** (-3.154)	-0.795 *** (-3.481)	-1.277 *** (-4.142)	-0.948 *** (-3.689)	-0.953 *** (-3.707)
R&D Expenditures	-0.007 (-1.477)	-0.007 * (-1.643)	-0.033 *** (-3.896)	-0.007 (-1.365)	-0.007 (-1.367)
Diversification	-0.095 *** (-4.168)	-0.096 *** (-4.677)	-0.041 *** (-3.189)	-0.096 *** (-4.067)	-0.095 *** (-4.007)
Industry Effect	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Random
Adjusted R-squared	0.739	0.739	0.924	0.915	0.915

Max VIF	1.362	1.362	1.362	1.362	1.362
Durbin Watson Statistic	2.118	2.118	1.949	1.947	1.947
Panel C: Impacts of CEO Age and CEO Tenure on Corporate Cash Holdings					
Obs. = 7,249	Panel Regression		Panel 2SLS Regression		
	Dependent variable = cash holdings		Dependent variable = cash holdings		
Model		1	2	3	4
CEO Characteristics					
Middle-Aged CEO	-0.040 (-0.159)				
Old CEO	-0.527 ** (-2.398)				
Medium CEO Tenure	0.115 (0.497)				
Long CEO Tenure	0.564 ** (2.407)				
CEO Age		-0.005 ** (-2.162)	0.033 *** (3.016)	-0.051 *** (-9.240)	0.060 ** (2.286)
(CEO Age) ²			-0.004 ** (-2.042)		-0.006 ** (-2.223)
CEO Tenure		0.014 *** (3.121)	0.026 *** (3.109)	0.048 *** (3.372)	1.355 *** (2.474)
(CEO Tenure) ²			-0.003 (-0.054)		-0.035 (-1.234)
Control Variables	Yes	Yes	Yes	Yes	Yes
Constant	Yes	No	No	Yes	N0
Industry Effect	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Random
Adjusted R-squared	0.778	0.792	0.806	0.809	0.639
Max VIF	1.407	1.407	1.407	1.407	1.407
Durbin Watson Statistic	2.094	1.964	1.949	2.058	1.965

Note. We use the national average life expectancy for the year each CEO is 30 years old, as the instrumental variable of CEO age, and the average tenure of CEOs in the same industry in the previous year as the instrumental variable of CEO tenure. *t* values are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2 Moderating effects of CEO internalization

In Table 4, we further examine the impact of CEO internalization on cash holdings in two parts: the original effect which is generated by CEO internalization dummy variable and the moderating effect which is measured by the interaction terms between the CEO internalization dummy and CEO age, (CEO Age) $\times$ (CEO Internalization), or CEO Tenure, (CEO Tenure) $\times$ (CEO Internalization)²².

Table 4 shows that CEO age is negatively correlated with cash holdings and CEO internalization has no significant effect on cash holdings on panel regression and panel 2SLS regression. However, the interaction terms between the CEO internalization dummy and CEO age positively affect cash holdings in Model 4 but not significantly in Model 1. The negative effects of CEO age (estimated coefficient is -0.114) on firm cash holdings are greater than the positive effects of the interaction terms between CEO internalization and CEO age (0.037) on firm cash holdings in Model 4, indicating that CEO internalization mitigates the negative effects of CEO age on firm cash holdings.

This empirical evidence indicates that internalized CEOs decrease the negative effect of CEO age on cash holdings, implying that internalized CEOs have fewer career concerns, lower signal effects, and lower resource dependence effects than non-internalized CEOs. Therefore, Hypothesis 3 is supported.

Models 2 and 5 in Table 4 support that the longer the CEO tenure, the greater the cash holdings, and the estimated coefficients for the interaction terms between CEO internalization and CEO tenure are positive. This implies that CEO internalization strengthens the positive effect of CEO tenure on cash holdings. Therefore, Hypothesis 4 is not supported.

Models 3 and 6 in Table 4 include both CEO age and CEO tenure variables in the regression models, and the

²² Table 4 focuses on the key factors of CEO age, CEO tenure, and both CEO age and CEO tenure, respectively, by controlling for other variables, as in Table 3.

empirical results are mostly consistent with the other models. We also observe that although CEO internalization does not affect firm cash holdings directly, it does affect them indirectly through the effects of CEO age and tenure. In addition, the estimated coefficients of the square term of CEO age are not significant in all models, indicating that there is no significant non-linear relationship between CEO age and corporate cash holdings, which differs from the results in Table 3.

Table 4

Moderating Effects of CEO Internalization on the Relationships between CEO Age or CEO Tenure and Corporate Cash Holdings

Obs. = 7,249 Model	Panel Regression			Panel 2SLS Regression		
	Dependent variable = cash holdings			Dependent variable = cash holdings		
	1	2	3	4	5	6
CEO Characteristics						
CEO Age	-0.013 ** (-2.094)		-0.107 *** (-3.033)	-0.114 *** (-2.541)		-0.023 *** (-4.306)
(CEO Age) ²	-0.002 (-1.027)		-0.001 (-0.348)	-0.002 (-1.344)		-0.002 * (-1.849)
CEO Tenure		0.028 ** (2.123)	0.025 *** (6.539)		0.023 *** (3.356)	0.081 *** (4.861)
CEO Internalization	0.170 (0.138)	0.100 (0.469)	0.041 (1.473)	0.119 (1.502)	0.056 (0.218)	0.028 (1.442)
Interactive Items						
CEO Age× CEO Internalization	0.003 (0.163)		0.022 *** (2.469)	0.037 *** (2.626)		0.011 *** (3.357)
CEO Tenure× CEO Internalization		0.016 *** (2.956)	0.016 ** (2.104)		0.010 *** (3.461)	0.008 ** (2.082)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	Yes	Yes	Yes	No
Industry Effect	Yes	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Fixed	Random
Adjusted R-squared	0.771	0.744	0.812	0.768	0.744	0.763
Max VIF	2.462	4.479	4.479	2.462	4.479	4.479
Durbin Watson Statistic	2.080	2.121	2.097	1.989	2.121	2.086

Note. *t* values are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3 Observation Period Effect of CEO Tenure

Following Intintoli and Kahle (2016), we establish a dummy variable observation period which is equal to 1 if the CEO's tenure in the current company is less than two years; otherwise, it is equal to 0. Table 5 shows that the observation period is greater than 0 and significant in all models (Models 1, 3, and 5 on panel regression and Models 6, 8, and 10 on panel 2SLS regression), which implies that CEOs increase cash holdings during the observation period. Thus, there is an observation period effect on the cash management practices of CEOs, which supports Hypothesis 5A.

Models 3-5 in Table 5 demonstrate the moderating effects of CEO internalization for the observation period by adding the interaction term (Observation Period) × (CEO Internalization) based on the model in Models 1 and 6. Our results show that the observation period positively impacts on cash holdings while CEO internalization has no impact on cash holdings for all models, which is similar to the results in Table 4. The estimated coefficients for the interaction term (Observation Period) × (CEO Internalization) are negative and significant in Models 3, 5, 8, and 10. This implies that CEO internalization has a moderating effect by decreasing the positive effect of the observation period on cash holdings. Our results support Hypothesis 5B.

4.4 Threshold Effect of CEO Tenure

To test for a threshold effect on CEO tenure, we establish a tenure threshold period dummy variable which is equal to 1 if the CEO's current tenure minus the target tenure (industry-averaged tenure) is less than two years and 0 otherwise (Hu, Kale, et al., 2011). The results of Models 2, 4, and 5 on panel regression and Models 7, 9, and 10 on panel 2SLS regression are shown in Table 6. Both panel regression and panel 2SLS regression methods show that the estimated coefficient of the threshold period is significant and positive, indicating that CEOs approaching the industry average tenure are more likely to increase cash holdings. Therefore, a tenure threshold effect exists, in terms of CEO cash management practices, which supports Hypothesis 6A.

Table 5
Observation Period Effect and Threshold Effect of CEO Tenure

Obs. = 7,249	Panel Regression					Panel 2SLS Regression				
	1	2	3	4	5	6	7	8	9	10
CEO Characteristics										
CEO Age	-0.012 ** (-2.057)	-0.009 ** (-2.001)	-0.025 ** (-2.117)	-0.013 ** (-2.013)	-0.016 *** (-3.857)	-0.006 *** (-3.487)	-0.019 *** (2.647)	-0.022 *** (-3.151)	-0.019 *** (-3.123)	-0.029 *** (-3.175)
(CEO Age) ²	-0.002 (-0.081)	-0.001 (-0.026)	-0.001 (-0.128)	-0.002 (-0.043)	-0.004 (-1.065)	-0.002 * (-1.815)	-0.001 * (-1.784)	-0.001 (-1.565)	-0.001 (-1.526)	-0.014 ** (-2.106)
CEO Tenure	0.008 ** (2.237)	0.003 ** (2.094)	0.011 *** (3.341)	0.011 ** (2.032)	0.015 *** (4.215)	0.007 *** (2.489)	0.009 *** (2.495)	0.019 *** (3.520)	0.013 *** (2.531)	0.011 ** (2.006)
CEO Internalization			0.057 (1.007)	0.066 (0.512)	0.058 (1.254)			0.076 (0.141)	0.068 (1.240)	0.032 (1.483)
Observation Period Effect										
Observation period	0.108 ** (2.240)		0.626 *** (3.061)		0.313 *** (4.372)	0.113 *** (4.372)		1.027 *** (2.707)		1.441 *** (5.127)
Threshold Effect										
Threshold Period		0.009 *** (3.541)		0.025 *** (4.038)	0.032 *** (2.955)		0.083 *** (2.832)		0.027 *** (3.061)	0.025 *** (3.105)
Moderating Effect										
Observation period × CEO Internalization			-0.410 ** (-2.013)		-0.168 ** (-2.210)			-0.779 *** (-2.458)		-0.439 *** (-4.111)
Threshold Period × CEO Internalization				-0.741 (-0.946)	-0.412 (-1.062)				-0.727 (-0.987)	-0.516 (-1.009)
CEO Age × CEO Internalization					0.009 * (1.922)					0.012 ** (2.204)
CEO Tenure × CEO Internalization					0.012 ** (2.112)					0.015 ** (1.991)
Control Variables										
Constant	No	No	No	No	No	No	No	No	No	No
Industry Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Fixed	Random	Random	Random	Random	Random
Adjusted R-squared	0.806	0.806	0.806	0.806	0.841	0.747	0.763	0.762	0.762	0.804
Max VIF	1.348	1.408	1.410	1.408	1.410	1.348	1.408	1.410	1.408	1.410
Durbin Watson Statistic	2.003	2.001	2.007	2.137	2.106	2.168	2.098	2.085	2.168	2.091

Note. the values in () represent the *t*-values of the estimated coefficients test, and *, **, *** indicate that the estimated values are significant at the 10%, 5%, and 1% levels, respectively.

Models 4-5 and 9-10 in Table 5 demonstrate the moderating effects of CEO internalization for the threshold period by adding the interaction term (Threshold Period) $\times$ (CEO Internalization) based on the model in Models 2 and 7. The results show that the threshold period has a positive impact on cash holdings and CEO internalization has no effect on cash holdings. The estimated coefficients for the interaction term (Threshold Period) $\times$ (CEO Internalization) are negative and insignificant in Models 4, 5, 9, and 10. Our results do not support that CEO internalization indirectly affects cash holdings through the threshold period effect, and Hypothesis 6B is not supported.

4.5 Robustness Test²³

Family control is a characteristic of listed companies in Taiwan. To determine whether family control affects the relationship between CEO age, tenure, and corporate cash holdings, this study divides the data into two groups: family firms and non-family firms, and conducts an empirical analysis. The results are presented in the Appendix (Table A1). The findings indicate that, for both family and non-family firms, CEO age has a negative relationship with corporate cash holdings, while CEO tenure has a positive relationship. These results are consistent with those from the overall sample. The only difference is that, after grouping, the nonlinear relationship between CEO age and corporate cash holdings no longer exists.

Lim and Lee (2019) use data from Korea to examine the impact of CEO characteristics on excess cash holdings. The CEO characteristics analyzed in their study include tenure, owner-manager status, stock options, and whether the CEO is part of a Korean chaebol group. However, CEO age is not included. Their results show a U-shaped relationship between CEO tenure and excess cash holdings.

This study adopts the model from Lim and Lee (2019) to estimate the ratio of excess cash holdings, using the ratio of excess cash holdings $\times$ 100 as the dependent variable, to examine the impact of CEO age and tenure on excess cash holdings. The empirical results are presented in the Appendix (Table A2). Our findings show that, except for the absence of a nonlinear relationship between CEO age and excess cash holdings when both CEO age and tenure are considered together, the results are consistent with those in Table 3 regarding the impact of CEO age and tenure on cash holdings.

This result also highlights the relationship between CEO age, tenure, and cash holdings, as well as agency problems in Taiwanese listed companies. However, unlike Lim and Lee (2019), this study finds a positive linear relationship between CEO tenure and excess cash holdings. These test results further demonstrate the robustness of this study's findings.

5. Conclusion

In this study, we analyze the impacts of CEO age and CEO tenure on cash holdings by controlling for standard firm characteristic variables and year fixed effects using individual CEO data from listed firms in Taiwan from 2010 to 2019. Our results show that CEO age is negatively correlated with cash holdings. This indicates that the negative effects of career concerns and resource dependence on the relationship between CEO age and cash holdings is greater than the positive effect of signal. The weaker effect of signal implies that the opportunity cost for young CEO to make high risk investment is relatively high in Taiwan. Thus, firms with younger CEO hold more cash. In addition, the test results for the non-linear relationship between CEO age and cash holdings show that although in Models 3 and 6 in Panel A of Table 3, CEO age and cash holdings have an inverted U relationship, the turning point of the relationship between CEO age and cash holdings from positive to negative occurs before the CEO is 10 years old, while the youngest CEO in the sample is 23 years old. Therefore, within the range of observations in this study, the relationship between CEO age and cash holdings is negative.

The impact of tenure on cash holdings mainly comes from accumulated work experience, CEO power, human capital, and overconfidence. With the accumulation of work experience during a CEO's tenure, cash holdings are reduced, while increased power and human capital lead to increased cash holdings. Although overconfidence produced by longer tenure may have a positive or negative impact on cash holdings, the relationship between tenure and cash holdings is determined by the sum of these four effects. The empirical results of this study show that CEO tenure has a positive association with firm cash holdings. This means that the sum of the effects of the enhancement of CEO power and the increase in human capital from longer CEO tenure is greater than the effect of the accumulation of work experience on firm cash holdings.

CEO internalization is a common phenomenon among Taiwan's listed companies. The estimated coefficient of CEO internalization on cash holdings is not significant; however, the interaction term between CEO internalization and CEO age is significant and positive, which implies that internalized CEOs differ from non-internalized CEOs in terms of fewer career concerns, weaker motivation for signaling of ability, and less resource dependency. CEO internalization mitigates the negative impact of CEO age on cash holdings. In addition, the interaction term between CEO internalization and CEO tenure is significant and positive, implying that internalized CEOs have less performance pressure, greater power, and more human capital. Increasing power and position increase the chance of grabbing

²³ We thank the referee for this suggestion.

shareholders' wealth and increasing human capital strengthens the motivation of a CEO to consolidate his/her position, leading to a conservative strategy. CEO internalization enhances the positive impact of CEO tenure on cash holdings. However, the effect of CEO internalization on cash holdings is mainly from the moderate effects of CEO age and tenure on cash holdings.

Our results show that CEOs maintain significantly higher cash holdings during the observation period and the target tenure threshold period, implying that the board expects a new CEO to bring about change to improve firm performance. In the case of information asymmetry, whether a new CEO's performance meets the board's expectations determines whether that CEO will be allowed to continue in his/her position. To pass assessment, a new CEO is expected to hold more cash and avoid uncertain risky investments. CEO tenure is often viewed by the labor market as an indicator of a CEO's management ability. Therefore, a CEO takes the industry average tenure as his/her target tenure threshold. This expectation is stronger as the tenure approaches the target threshold. Therefore, cash holdings are expected to increase to hedge against uncertain risks and allow a CEO to exceed the target tenure threshold.

In addition, we find that the estimated coefficient of the interaction term between CEO internalization and observation period is significant and negative, but the estimated coefficient of the interaction term between CEO internalization and threshold period is negative and insignificant. These results imply that an internalized CEO, who is also a controlling shareholder, is less likely to be penalized for bad decisions or underperforming than a professional CEO and experiences less pressure during the observation period, thereby reducing the effect on cash holdings during the observation period. Although internalized CEOs have less pressure to perform than non-internalized CEOs, they still desire to have their management skills affirmed by the labor market.

Finally, our empirical results provide useful implications for the CEO labor market and government policies for developing a corporate governance system. Our empirical results show that CEO personal characteristics, especially age and tenure, are very important factors for determining the level of corporate cash holdings mainly due to risk attitudes and self-interest. If a company with controlling shareholders hires a professional manager as CEO, the shareholders with absolute control can effectively monitor the professional CEO, reducing the increase in cash holdings due to conflict between managers and shareholders. However, most CEOs of listed companies in Taiwan are controlling shareholders. Thus, we suggest that the authorities strengthen investor protection by establishing an external oversight mechanism and a more stringent corporate governance system.

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Constant	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Industry Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
Adjusted <i>R</i> -squared	0.745	0.735	0.745	0.735	0.745	0.735	0.745	0.735	0.696	0.648
Max VIF	1.362	1.843	1.362	1.843	1.362	1.843	1.362	1.843	1.362	1.843
Durbin Watson Statistic	2.215	2.116	2.148	2.121	2.067	2.118	2.098	2.068	1.925	1.906
Observations	5002	2247	5002	2247	5002	2247	5002	2247	5002	2247

Panel C: Impacts of CEO Age and CEO Tenure on Corporate Cash Holdings (Y = Cash Holdings)

Model	Panel Regression				Panel 2SLS Regression					
	1		2		3		4		5	
	Family	Non-Family	Family	Non-Family	Family	Non-Family	Family	Non-Family	Family	Non-Family
CEO Characteristics										
Middle-Aged CEO	0.215 (0.032)	0.271 (1.485)								
Old CEO	-0.127 * (-1.875)	-0.251 ** (-2.471)								
Medium CEO Tenure	0.319 (2.283) **	0.233 *** (2.596)								
Long CEO Tenure	-0.020 (-0.486)	-0.015 (-1.255)								
CEO Age			-0.017 *** (-2.573)	-0.066 * (-1.939)	-0.101 ** (-0.772)	-0.185 * (-1.883)	-0.019 ** (-2.102)	-0.105 *** (-2.742)	-0.171 ** (-2.224)	-0.238 *** (-2.614)
(CEO Age) ²					0.002 (0.670)	0.008 (1.069)			0.002 (0.234)	0.015 (1.264)
CEO Tenure			0.058 ** (2.283)	0.046 *** (2.763)	0.165 ** (2.296)	0.154 ** (2.181)	0.053 ** (2.182)	0.034 * (1.899)	0.116 ** (2.293)	0.065 ** (2.302)
(CEO Tenure) ²					-0.001 (-0.408)	-0.005 (-1.105)			-0.011 (-0.339)	-0.002 (-0.043)
Control Variables										
Constant	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Effect	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No	No
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
Adjusted <i>R</i> -squared	0.780	0.762	0.780	0.762	0.780	0.762	0.776	0.760	0.745	0.724
Max VIF	1.407	1.967	1.407	1.967	1.407	1.967	1.407	1.967	1.407	1.967
Durbin Watson Statistic	2.110	2.162	2.106	2.188	2.106	2.162	2.063	2.145	1.973	1.927
Observations	5002	2247	5002	2247	5002	2247	5002	2247	5002	2247

Note. We use the national average life expectancy for the year each CEO is 30 years old, as the instrumental variable of CEO age, and the average tenure of CEOs in the same industry in the previous year as the instrumental variable of CEO tenure. *t* values are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2***The Impact of CEO Age and Tenure on Excess Cash Holdings*****Panel A: Impact of CEO Age on Excess Cash Holdings (Y = Excess Cash Holdings)**

Obs.=7,249	Panel Regression			Panel 2SLS Regression	
Model	1	2	3	4	5
CEO Characteristics					
Middle-Aged CEO	0.577 (0.941)				
Old CEO	-0.510 ** (-2.059)				
CEO Age		-0.009 ** (-2.421)	0.065 ** (2.128)	-0.032 ** (-2.103)	0.046 * (1.704)
(CEO Age) ²			-0.021 * (-2.222)		-0.022 * (-1.685)
Control Variables	Yes	Yes	Yes	Yes	Yes
Excess Cash Holdings (-1)	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	No	No	No
Industry Effect	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Fixed
Adjusted R-squared	0.786	0.786	0.786	0.790	0.790
Max VIF	1.404	1.406	1.406	1.406	1.406
Durbin Watson Statistic	1.975	1.958	1.958	1.905	1.905

Panel B: Impact of CEO Tenure on Excess Cash Holdings ((Y = Excess Cash Holdings))

Obs.=7,249	Panel Regression			Panel 2SLS Regression	
Model	1	2	3	4	5
CEO Characteristics					
Medium CEO Tenure	0.204 *** (2.828)				
Long CEO Tenure	-0.009 (-0.316)				
CEO Tenure		0.050 *** (3.345)	0.138 *** (2.674)	0.030 ** (2.424)	0.050 ** (2.256)
(CEO Tenure) ²			-0.001 (-1.599)		-0.001 (-1.191)
Control Variables	Yes	Yes	Yes	Yes	Yes
Excess Cash Holdings (-1)	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	No	Yes	No
Industry Effect	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Fixed
Adjusted R-squared	0.730	0.730	0.730	0.729	0.715
Max VIF	1.362	1.362	1.362	1.362	1.362
Durbin Watson Statistic	2.063	2.063	2.063	2.061	1.960

Panel C: Impacts of CEO Age and CEO Tenure on Excess Cash Holdings ((Y = Excess Cash Holdings))

Obs.=7,249	Panel Regression			Panel 2SLS Regression	
Model	1	2	3	4	5
CEO Characteristics					
Middle-Aged CEO	0.430 (0.694)				
Old CEO	-0.159 ** (-2.255)				
Medium CEO Tenure	0.261 ** (2.422)				
Long CEO Tenure	-0.031 (-0.816)				
CEO Age		-0.022 *	-0.118 *	-0.039 *	-0.095 **

		(-1.931)	(-1.789)	(-1.943)	(-2.492)
(CEO Age) ²			0.001		0.003
			(0.946)		(0.589)
CEO Tenure		0.040 *	0.145 **	0.081 *	0.072 **
		(1.837)	(2.195)	(1.848)	(2.061)
(CEO Tenure) ²			-0.003		-0.035
			(-1.538)		(-1.081)
Control Variables	Yes	Yes	Yes	Yes	Yes
Excess Cash Holdings (-1)	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	No	No	No
Industry Effect	Yes	Yes	Yes	Yes	Yes
Cross-Section Effect	Fixed	Fixed	Fixed	Fixed	Fixed
Adjusted R-squared	0.787	0.786	0.786	0.789	0.756
Max VIF	1.407	1.407	1.407	1.407	1.407
Durbin Watson Statistic	1.958	1.956	1.957	1.997	1.982

Note. We use the national average life expectancy for the year each CEO is 30 years old, as the instrumental variable of CEO age, and the average tenure of CEOs in the same industry in the previous year as the instrumental variable of CEO tenure. *t* values are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.